

Plum Claims Evaluation Report

Google Docs ready formatted version of the policy pipeline evaluation

Prepared from generated evaluation artifacts on 2026-09-27.

Executive Result

Conclusion: 12 of 12 supplied structured fixture cases matched the expected decision, amount, reason, confidence rule, and explicit behavior checks.

Scope: This evaluation proves deterministic policy pipeline behavior against structured fixtures. It does not prove OCR accuracy, handwriting support, multilingual extraction, image quality handling, or calibrated confidence.

Field	Value
Policy	PLUM_GHI_2024
Cases	12
Result	12/12 matched
Policy file sha256	1b19689948d8273c32ec2b5f35c75c25ad48a5ae46b937092c464178de4484ce
Canonical policy sha256	d07403592834f8d1011398ac62607c5c423a3da4ab2aa1ea347b8554394a34e4
Fixture file sha256	4b9b9a047ec6a6479a81f6b2767f00f931920ad7bedb3f547abb54b771034e63
Full machine output	docs/reports/evaluation-data.json

Case Results

Case	Scenario	Expected	Produced	Amount	Match
TC001	Wrong Document Uploaded	None	None	None	Yes
TC002	Unreadable Document	None	None	None	Yes
TC003	Documents Belong to Different Patients	None	None	None	Yes
TC004	Clean Consultation – Full Approval	Approved	Approved	Rs 1,350	Yes
TC005	Waiting Period – Diabetes	Rejected	Rejected	Rs 0	Yes
TC006	Dental Partial Approval – Cosmetic Exclusion	Partial	Partial	Rs 8,000	Yes
TC007	MRI Without Pre-Auth	Rejected	Rejected	Rs 0	Yes
TC008	Per-Claim Limit Exceeded	Rejected	Rejected	Rs 0	Yes

Case	Scenario	Expected	Produced	Amount	Match
TC009	Fraud Signal — Same-Day Claims	Review	Review	Rs 0	Yes
TC010	Network Hospital — Discount Applied	Approved	Approved	Rs 3,240	Yes
TC011	Component Failure — Graceful Degradation	Approved	Approved	Rs 4,000	Yes
TC012	Excluded Treatment	Rejected	Rejected	Rs 0	Yes

Interpretation and Limits

- The engine reads only the canonical policy produced from the unmodified policy file; every interpretation, merge, and conflict resolution is captured in the canonical audit trail and referenced from decision traces.
- The per-claim ceiling is the maximum of the global per-claim limit and the relevant category sub-limit, tested on eligible amount after excluded lines are removed. A matched pre-authorization rule governs amounts above the ceiling instead.
- Category sub-limits are annual per-member caps on net benefit for the category's own service lines. For consultation, tests and medicines billed with a consultation are not treated as consultation service lines.
- Aggregate limits are applied when utilisation accompanies the claim. If utilisation is absent, those rules are recorded as not evaluated, disclosed as advisory reasons on payable outcomes, and reflected in confidence.
- The supplied cases do not carry a submission date, so the 30-day submission deadline is not evaluated for them. Web intake stamps a server-side submission date at intake.
- Confidence values are heuristic evidence-completeness scores, not calibrated probabilities. TC011 intentionally simulates optional risk enrichment failure and records the degraded confidence and review recommendation.

Case Output Summaries

TC001 Wrong Document Uploaded

Result: NEEDS_CORRECTION; approved amount not applicable; match yes.

Primary evidence signal: DOCUMENT_MISSING

Explicit behavior checks: passed

TC002 Unreadable Document

Result: NEEDS_CORRECTION; approved amount not applicable; match yes.

Primary evidence signal: DOCUMENT_UNREADABLE

Explicit behavior checks: passed

TC003 Documents Belong to Different Patients

Result: NEEDS_CORRECTION; approved amount not applicable; match yes.

Primary evidence signal: PATIENT_MISMATCH

Explicit behavior checks: passed

TC004 Clean Consultation — Full Approval

Result: APPROVED; approved amount Rs 1,350; match yes.

Primary evidence signal: CATEGORY_SUB_LIMIT_HISTORY_NOT_EVALUATED

Explicit behavior checks: passed

TC005 Waiting Period — Diabetes

Result: REJECTED; approved amount Rs 0; match yes.

Primary evidence signal: WAITING_PERIOD

Explicit behavior checks: passed

TC006 Dental Partial Approval — Cosmetic Exclusion

Result: PARTIAL; approved amount Rs 8,000; match yes.

Primary evidence signal: EXCLUDED_PROCEDURE; TREATMENT_DATE_NOT_CORROBORATED;
ADVISORY_DOCUMENT_ABSENT; CATEGORY_SUB_LIMIT_HISTORY_NOT_EVALUATED

Explicit behavior checks: passed

TC007 MRI Without Pre-Authorization

Result: REJECTED; approved amount Rs 0; match yes.

Primary evidence signal: PRE_AUTH_MISSING

Explicit behavior checks: passed

TC008 Per-Claim Limit Exceeded

Result: REJECTED; approved amount Rs 0; match yes.

Primary evidence signal: PER_CLAIM_EXCEEDED

Explicit behavior checks: passed

TC009 Fraud Signal — Multiple Same-Day Claims

Result: MANUAL_REVIEW; approved amount Rs 0; match yes.

Primary evidence signal: SAME_DAY_CLAIMS; CATEGORY_SUB_LIMIT_UNVERIFIED

Explicit behavior checks: passed

TC010 Network Hospital — Discount Applied

Result: APPROVED; approved amount Rs 3,240; match yes.

Primary evidence signal: TREATMENT_DATE_NOT_CORROBORATED;
CATEGORY_SUB_LIMIT_HISTORY_NOT_EVALUATED

Explicit behavior checks: passed

TC011 Component Failure — Graceful Degradation

Result: APPROVED; approved amount Rs 4,000; match yes.

Primary evidence signal: COMPONENT_DEGRADED; PATIENT_IDENTITY_NOT_VERIFIED; TREATMENT_DATE_NOT_CORROBORATED; SESSION_HISTORY_NOT_EVALUATED

Explicit behavior checks: passed

TC012 Excluded Treatment

Result: REJECTED; approved amount Rs 0; match yes.

Primary evidence signal: EXCLUDED_CONDITION; WAITING_PERIOD; EXCLUDED_PROCEDURE

Explicit behavior checks: passed

Generated Source Files

- docs/reports/evaluation.md contains the complete Markdown report with trace excerpts.
- docs/reports/evaluation-data.json contains the full structured output for every case.
- docs/reports/verification-summary.md contains the overall verification summary generated by make verify.